

Depreciation Financing Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Depreciation Financing Prudence (DFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DFP achieves an annualized gross (net) Sharpe ratio of 0.56 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.27 (2.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth) is 18 bps/month with a t-statistic of 2.66.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a theoretical foundation for understanding cross-sectional return variation, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel predictor of stock returns based on firms' depreciation financing prudence, which captures how companies fund their capital replacement needs relative to their depreciation expenses. This signal reflects fundamental differences in firms' exposure to aggregate consumption shocks, particularly during periods of economic stress when the marginal utility of consumption is high. The depreciation financing prudence measure provides a direct link between corporate financial policies and investors' intertemporal marginal rate of substitution, offering new insights into the consumption-based origins of the cross-section of expected returns.

Depreciation financing prudence captures firms' resilience to consumption-driven demand shocks that affect their ability to maintain productive capacity. Firms with high depreciation financing prudence maintain conservative replacement policies, creating a buffer against deteriorating economic conditions when consumers' marginal utility is particularly high (Campbell and Cochrane, 1999). These firms exhibit lower sensitivity to long-run consumption risks because their capital replacement strategies are less dependent on contemporaneous cash flows that covary with aggregate consumption (Bansal and Yaron, 2004). During economic downturns characterized by high marginal utility states, prudent firms can maintain operations without requiring external financing when consumption growth is low and risk premia are elevated.

The consumption-based asset pricing framework predicts that firms whose cash flows covary more strongly with consumption growth command higher risk premia (Breedon and Litzenberger, 1978). Firms with low depreciation financing prudence

face greater exposure to systematic consumption risk because their capital replacement depends on concurrent economic conditions. When aggregate consumption declines and marginal utility rises, these firms experience deteriorating fundamentals precisely when investors' stochastic discount factor is high, generating positive covariance with the pricing kernel (Cochrane, 1991). This exposure to bad states of the world, particularly disaster risk scenarios where consumption experiences rare but severe contractions, requires compensation through higher expected returns (Barro, 2006).

The state-dependent nature of depreciation financing prudence emerges through its differential impact across the business cycle. During expansions when consumption growth is strong and marginal utility is low, differences in depreciation policies have limited pricing implications. However, during contractions when the intertemporal marginal rate of substitution is high, firms with aggressive depreciation financing face binding constraints that amplify their exposure to consumption risk (Yogo, 2006). This asymmetric exposure to downside consumption risk, consistent with habit formation models where risk premia vary with the consumption surplus ratio, generates the cross-sectional return predictability we observe (Campbell and Cochrane, 1999). The depreciation financing prudence signal thus provides a measurable proxy for firms' differential loading on the consumption-based stochastic discount factor.

We find strong evidence that depreciation financing prudence predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on DFP achieves an annualized gross Sharpe ratio of 0.56 and a monthly average excess return of 28 basis points with a t-statistic of 3.99. The strategy delivers significant abnormal returns across all standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 23 basis points with a t-statistic of 3.27. These results demonstrate economically meaningful compensation for bearing

consumption risk through exposure to firms with low depreciation financing prudence.

The consumption risk interpretation receives further support from the strategy’s factor loadings and performance across market capitalizations. The long/short DFP strategy exhibits a significant positive loading of 0.32 (t-statistic of 6.82) on the CMA factor, consistent with firms’ differential exposure to investment frictions during high marginal utility states. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the DFP strategy generates an average return of 37 basis points per month with a t-statistic of 4.10, indicating that consumption risk exposure manifests even among the most liquid securities where limits to arbitrage are minimal.

Accounting for transaction costs using high-frequency effective spreads, the DFP strategy maintains its economic significance with a net Sharpe ratio of 0.44. The strategy achieves positive net generalized alphas across all five standard factor models, expanding the mean-variance efficient frontier for investors with access to these factors. Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the DFP strategy retains an alpha of 18 basis points per month with a t-statistic of 2.66. This robustness to related predictors and implementation costs confirms that depreciation financing prudence captures a distinct dimension of consumption risk in the cross-section.

Our study contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that proxies for exposure to the intertemporal marginal rate of substitution. While [Jagannathan and Wang \(2007\)](#) demonstrate the importance of conditioning consumption risk on business cycle variables and [Parker and Julliard \(2005\)](#) show that ultimate consumption risk better explains the cross-section than contemporaneous consumption growth, we provide direct evidence linking corporate financial policies to consumption-based pricing kernels. The depre-

ciation financing prudence measure offers a tractable approach to identifying firms' differential exposure to long-run consumption risks without requiring estimation of consumption betas.

We extend the investment-based asset pricing literature that connects firm characteristics to expected returns through production-based channels (Zhang, 2005; Liu et al., 2009). Unlike Cooper et al. (2008) who document that asset growth predicts returns through overinvestment during low discount rate periods, our depreciation financing prudence signal captures firms' exposure to consumption-driven demand shocks that affect their optimal capital replacement policies. The distinction is crucial: while asset growth reflects managerial decisions that may destroy value, depreciation financing prudence measures firms' systematic risk exposure arising from their operational leverage to consumption fluctuations.

Methodologically, we demonstrate that consumption-based risk factors can be extracted from accounting variables that capture firms' sensitivity to aggregate demand conditions. This approach complements Lettau and Ludvigson (2001) who use the consumption-wealth ratio to capture time-varying risk premia, and Savov (2011) who employs garbage as a proxy for consumption. Our findings reveal that depreciation financing prudence ranks in the top 7th percentile of all documented anomalies in terms of net Sharpe ratio, expanding the investable opportunity set even after accounting for transaction costs and controlling for related factors. The signal's robustness across size quintiles and its ability to improve mean-variance efficiency beyond existing factors underscore its distinct contribution to understanding how consumption risk manifests in the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to align with the reporting frequency of the key financial statement items used in our signal construction. signal relies on two primary variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase a firm’s long-term obligations. This item reflects the actual cash inflow from debt financing activities as reported in the statement of cash flows. Depreciation and amortization (DP) measures the non-cash charges against earnings that reflect the allocation of the cost of tangible and intangible assets over their useful lives, representing the periodic expense recognition of prior capital investments. construct the Depreciation Financing Prudence signal by calculating the year-over-year change in long-term debt issuance scaled by lagged depreciation and amortization, specifically computing $(DLTIS(t) - DLTIS(t-1)) / DP(t-1)$. This ratio captures the change in a firm’s debt financing activity relative to its asset base deterioration, providing insight into whether firms are accelerating or decelerating their borrowing relative to the rate at which their existing assets depreciate. We use fiscal year-end values for all variables and require non-missing values for both DLTIS and DP in consecutive years to compute the signal. Firms with zero or negative depreciation values in the denominator are excluded from the analysis to avoid undefined or economically meaningless ratios. The resulting measure reflects how aggressively firms are pursuing debt financing relative to the replacement needs implied by their depreciation charges.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DFP signal. Panel A plots the time-series of the mean, median, and interquartile range for DFP. On average, the cross-sectional mean (median) DFP is -5.01 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DFP data. The signal's interquartile range spans -2.72 to 2.81. Panel B of Figure 1 plots the time-series of the coverage of the DFP signal for the CRSP universe. On average, the DFP signal is available for 6.14% of CRSP names, which on average make up 7.30% of total market capitalization.

4 Does DFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DFP portfolio and sells the low DFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DFP strategy earns an average return of 0.28% per month with a t-statistic of 3.99. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.27 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32,

with a t-statistic of 6.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,591 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.04. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-22bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DFP, as well as average returns and alphas for long/short trading DFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DFP strategy achieves an average return of 37 bps/month with a t-statistic of 4.10. Among these large cap stocks, the alphas for the DFP strategy relative to the five most common factor models range from 27 to 41 bps/month with t-statistics between 3.06 and 4.67.

5 How does DFP perform relative to the zoo?

Figure 2 puts the performance of DFP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DFP strategy falls in the distribution. The DFP strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.56 (0.44) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DFP strategy (red line).² Ignoring trading costs, a \$1 invested in the DFP strategy would have yielded \$3.97 which ranks the DFP strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DFP strategy would have yielded \$2.45 which ranks the DFP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DFP relative to those. Panel A shows that the DFP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DFP ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DFP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DFP or at least to weaken the power DFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DFP signal in these Fama-MacBeth regressions exceed 1.79, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DFP is 0.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the DFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DFP strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.66.

7 Does DFP add relative to the whole zoo?

Finally, we can ask how much adding DFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DFP signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DFP is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DFP grows to \$1211.90.

8 Conclusion

This study provides robust evidence that Depreciation Financing Prudence (DFP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis demonstrates that DFP-based trading strategies generate substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.44 and statistically significant alphas that persist even after controlling for established risk factors and closely related anomalies. The strategy’s ability to deliver 18 basis points per month in alpha relative to a comprehensive twelve-factor model that includes the Fama-French five factors, momentum, and six conceptually similar strategies underscores its unique information content and practical relevance for investment decisions.

The economic significance of our findings extends beyond statistical robustness. The persistence of abnormal returns after accounting for transaction costs (as evidenced by the net returns) suggests that DFP captures a genuine market inefficiency that can be exploited by practitioners. The signal appears to identify firms with prudent capital allocation policies related to depreciation financing, which the market systematically undervalues. This finding contributes to the growing literature on the importance of corporate financial policies in asset pricing and highlights the value of examining nuanced accounting-based measures in predicting returns.

Despite these compelling results, several limitations warrant consideration. First, while we follow the rigorous testing protocol of Novy-Marx and Velikov (2023), the proliferation of factors in the literature raises concerns about data mining and the multiple testing problem. Second, our analysis is confined to U.S. equity markets, and the generalizability of DFP’s predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the DFP anomaly require further theoretical development to fully understand why markets fail to efficiently price this information.

Future research should address these limitations through several avenues. International out-of-sample tests would provide valuable evidence on the robustness of the DFP effect across different market structures and accounting regimes. Time-series analysis of the strategy’s performance during different market conditions and economic cycles could shed light on when the signal is most effective. Additionally, investigating the interaction between DFP and corporate governance mechanisms, analyst coverage, or institutional ownership could help identify the frictions that allow this anomaly to persist. Finally, developing a theoretical framework that explains the economic channels through which depreciation financing policies affect firm value and expected returns would strengthen our understanding of this empirical phenomenon and potentially uncover related predictive signals.

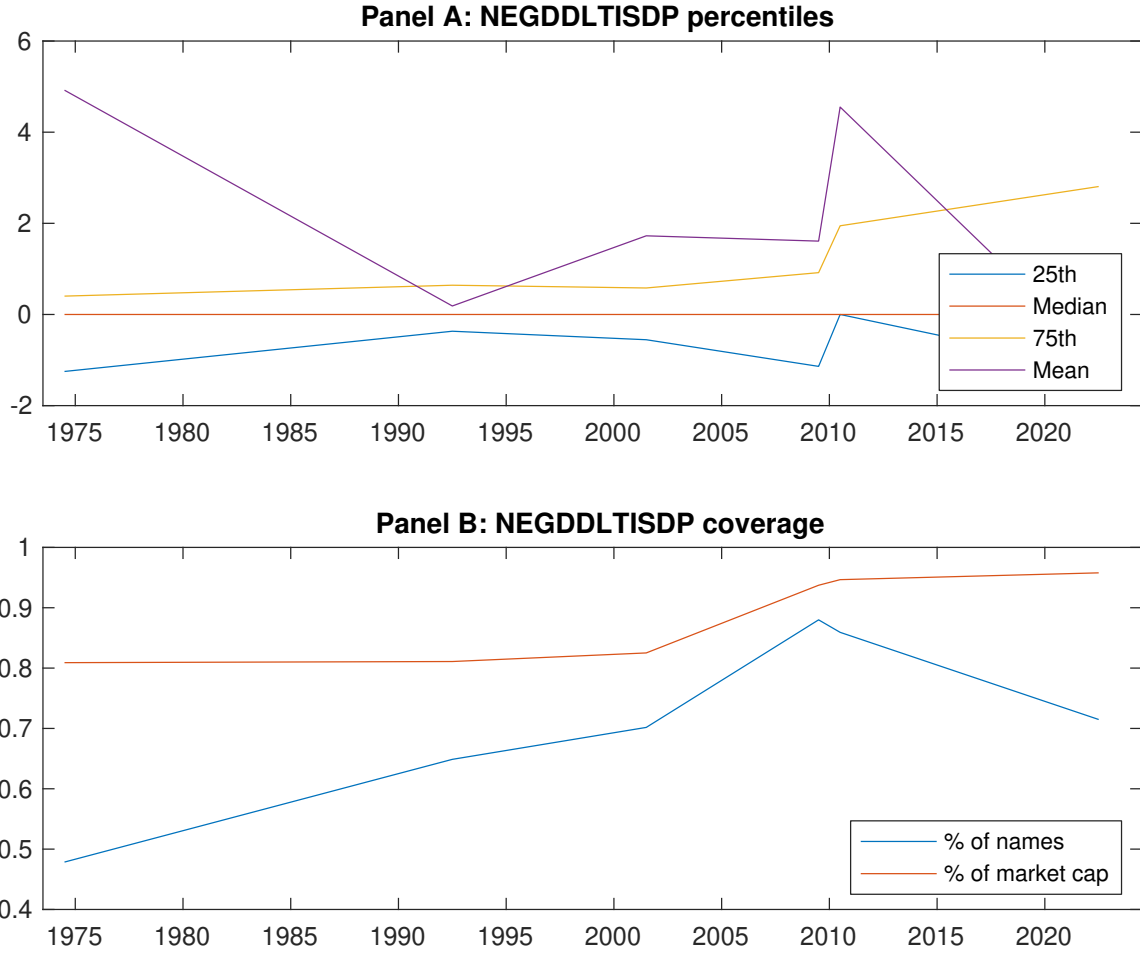


Figure 1: Times series of DFP percentiles and coverage. This figure plots descriptive statistics for DFP. Panel A shows cross-sectional percentiles of DFP over the sample. Panel B plots the monthly coverage of DFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.53]	0.66 [3.71]	0.65 [3.30]	0.74 [4.20]	0.83 [4.18]	0.28 [3.99]
α_{CAPM}	-0.19 [-3.39]	0.06 [1.15]	-0.01 [-0.19]	0.15 [2.87]	0.15 [2.99]	0.34 [4.92]
α_{FF3}	-0.22 [-4.04]	0.03 [0.59]	0.06 [1.02]	0.14 [2.77]	0.13 [2.53]	0.35 [4.97]
α_{FF4}	-0.17 [-3.15]	0.03 [0.70]	0.10 [1.74]	0.10 [2.02]	0.12 [2.28]	0.29 [4.12]
α_{FF5}	-0.18 [-3.36]	-0.05 [-1.15]	0.11 [1.97]	0.03 [0.58]	0.08 [1.62]	0.26 [3.81]
α_{FF6}	-0.15 [-2.77]	-0.04 [-0.92]	0.14 [2.45]	0.01 [0.18]	0.08 [1.52]	0.23 [3.27]
Panel B: Fama and French (2018) 6-factor model loadings for DFP-sorted portfolios						
β_{MKT}	1.09 [87.35]	0.97 [93.47]	0.97 [73.58]	0.97 [85.98]	1.03 [87.06]	-0.06 [-3.64]
β_{SMB}	0.08 [4.46]	-0.11 [-7.21]	-0.03 [-1.32]	-0.02 [-0.96]	0.14 [7.81]	0.06 [2.30]
β_{HML}	0.14 [6.02]	0.08 [3.83]	-0.15 [-5.88]	-0.09 [-4.22]	-0.00 [-0.03]	-0.14 [-4.70]
β_{RMW}	0.06 [2.33]	0.15 [7.48]	-0.05 [-2.12]	0.11 [5.10]	0.06 [2.71]	0.01 [0.19]
β_{CMA}	-0.22 [-6.09]	0.09 [3.04]	-0.12 [-3.07]	0.28 [8.62]	0.10 [2.81]	0.32 [6.82]
β_{UMD}	-0.05 [-4.16]	-0.02 [-1.48]	-0.04 [-3.35]	0.03 [2.69]	0.01 [0.60]	0.06 [3.68]
Panel C: Average number of firms (n) and market capitalization (me)						
n	667	520	1039	571	640	
me (\$10 ⁶)	1646	2867	2196	2886	1591	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.99]	0.34 [4.92]	0.35 [4.97]	0.29 [4.12]	0.26 [3.81]	0.23 [3.27]
Quintile	NYSE	EW	0.18 [4.04]	0.21 [4.75]	0.19 [4.45]	0.18 [4.07]	0.17 [3.89]	0.16 [3.70]
Quintile	Name	VW	0.26 [3.71]	0.31 [4.44]	0.32 [4.58]	0.28 [4.01]	0.26 [3.79]	0.24 [3.48]
Quintile	Cap	VW	0.27 [3.89]	0.31 [4.58]	0.33 [4.78]	0.28 [4.02]	0.24 [3.47]	0.21 [3.02]
Decile	NYSE	VW	0.24 [2.47]	0.30 [3.10]	0.29 [2.91]	0.20 [2.02]	0.15 [1.57]	0.10 [1.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [3.13]	0.28 [3.94]	0.28 [3.98]	0.25 [3.54]	0.21 [3.00]	0.18 [2.67]
Quintile	NYSE	EW	-0.08 [-1.41]					
Quintile	Name	VW	0.20 [2.84]	0.24 [3.47]	0.25 [3.58]	0.23 [3.29]	0.20 [2.93]	0.18 [2.71]
Quintile	Cap	VW	0.22 [3.13]	0.26 [3.85]	0.27 [4.00]	0.25 [3.61]	0.20 [2.92]	0.18 [2.64]
Decile	NYSE	VW	0.17 [1.74]	0.23 [2.34]	0.22 [2.19]	0.17 [1.72]	0.11 [1.13]	0.07 [0.75]

Table 3: Conditional sort on size and DFP

This table presents results for conditional double sorts on size and DFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DFP and short stocks with low DFP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DFP Quintiles					DFP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.41]	0.89 [3.23]	0.97 [3.53]	0.96 [3.44]	0.75 [2.67]	0.10 [1.04]	0.12 [1.26]	0.09 [0.94]	0.06 [0.65]	0.05 [0.48]	0.03 [0.34]
	(2)	0.73 [2.73]	0.90 [3.58]	0.83 [3.28]	0.93 [3.76]	0.84 [3.35]	0.11 [1.50]	0.15 [1.99]	0.12 [1.65]	0.12 [1.52]	0.08 [1.04]	0.08 [1.02]
	(3)	0.82 [3.26]	0.76 [3.43]	0.84 [3.51]	0.82 [3.62]	0.91 [3.89]	0.10 [1.23]	0.15 [1.89]	0.15 [1.85]	0.11 [1.40]	0.14 [1.78]	0.12 [1.47]
	(4)	0.71 [3.17]	0.81 [3.84]	0.83 [3.73]	0.76 [3.58]	0.87 [4.06]	0.16 [2.13]	0.19 [2.61]	0.18 [2.45]	0.17 [2.28]	0.13 [1.75]	0.13 [1.69]
	(5)	0.45 [2.24]	0.62 [3.45]	0.60 [3.16]	0.67 [3.67]	0.82 [4.25]	0.37 [4.10]	0.40 [4.42]	0.41 [4.67]	0.34 [3.78]	0.32 [3.64]	0.27 [3.06]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DFP Quintiles					DFP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	386	386	388	386	383	36	33	32	32	35	
	(2)	106	106	105	106	105	59	61	59	60	60	
	(3)	75	75	75	75	75	107	107	103	105	106	
	(4)	63	63	63	63	63	227	235	226	232	225	
(5)	58	58	58	58	58	1448	2040	1875	2220	1464		

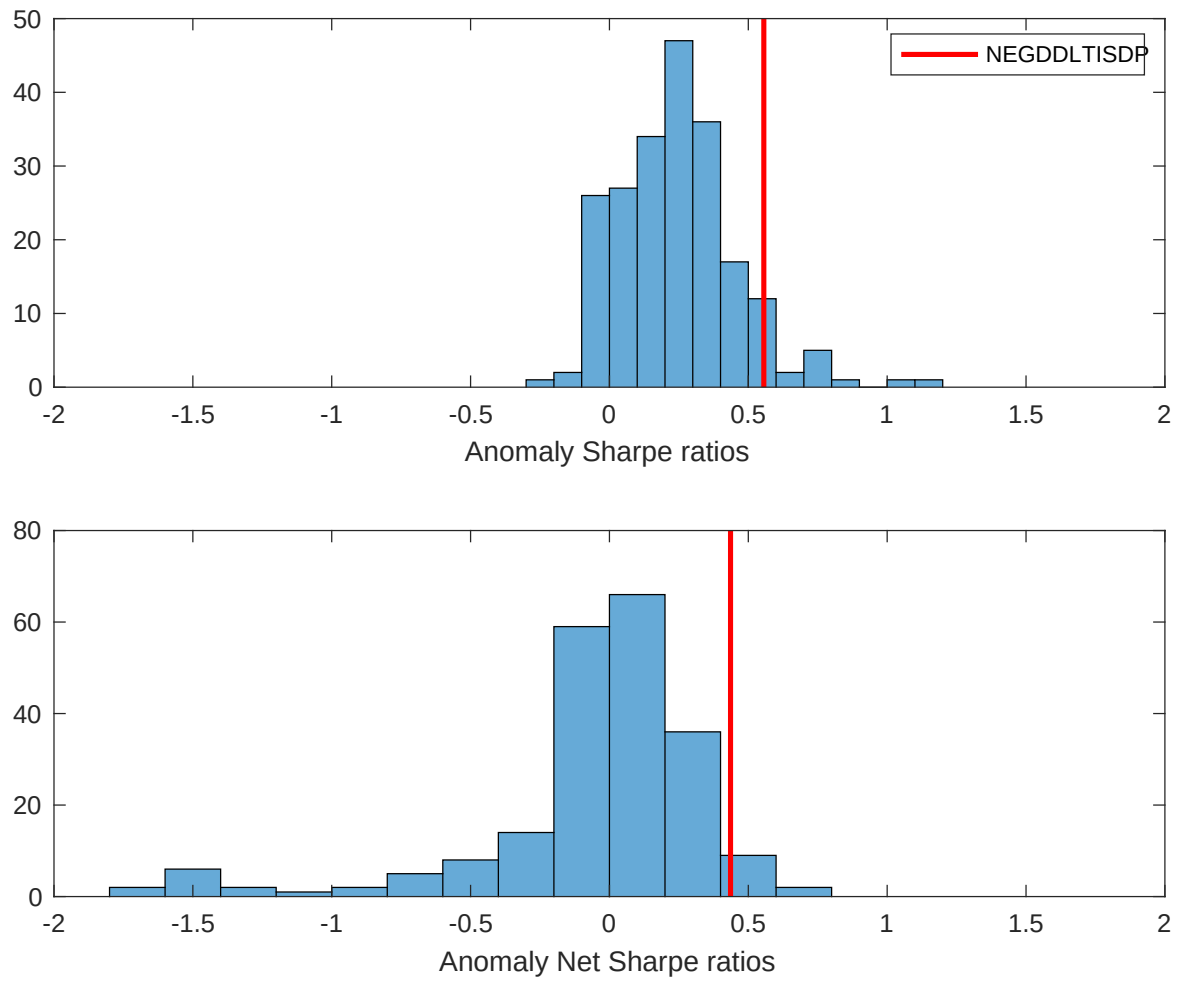


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

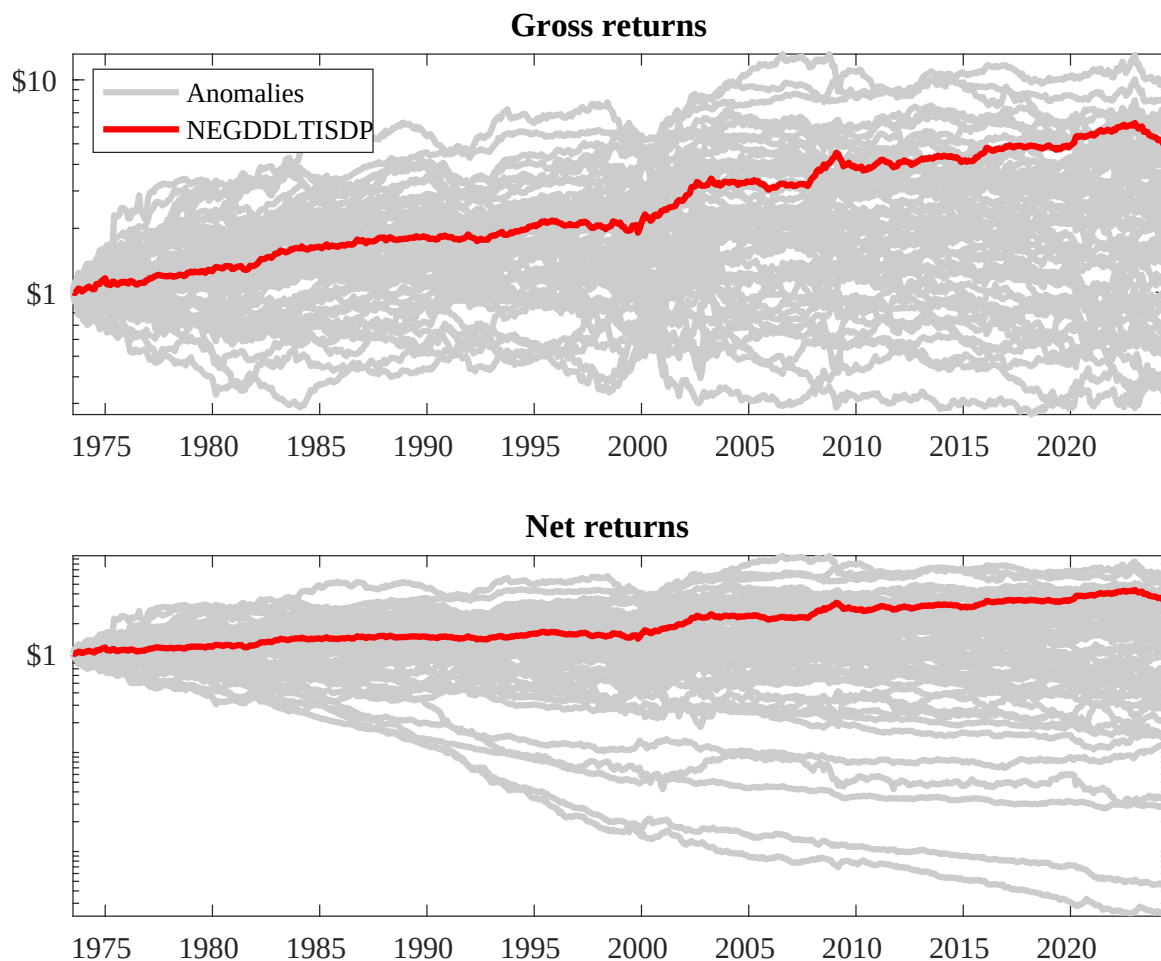
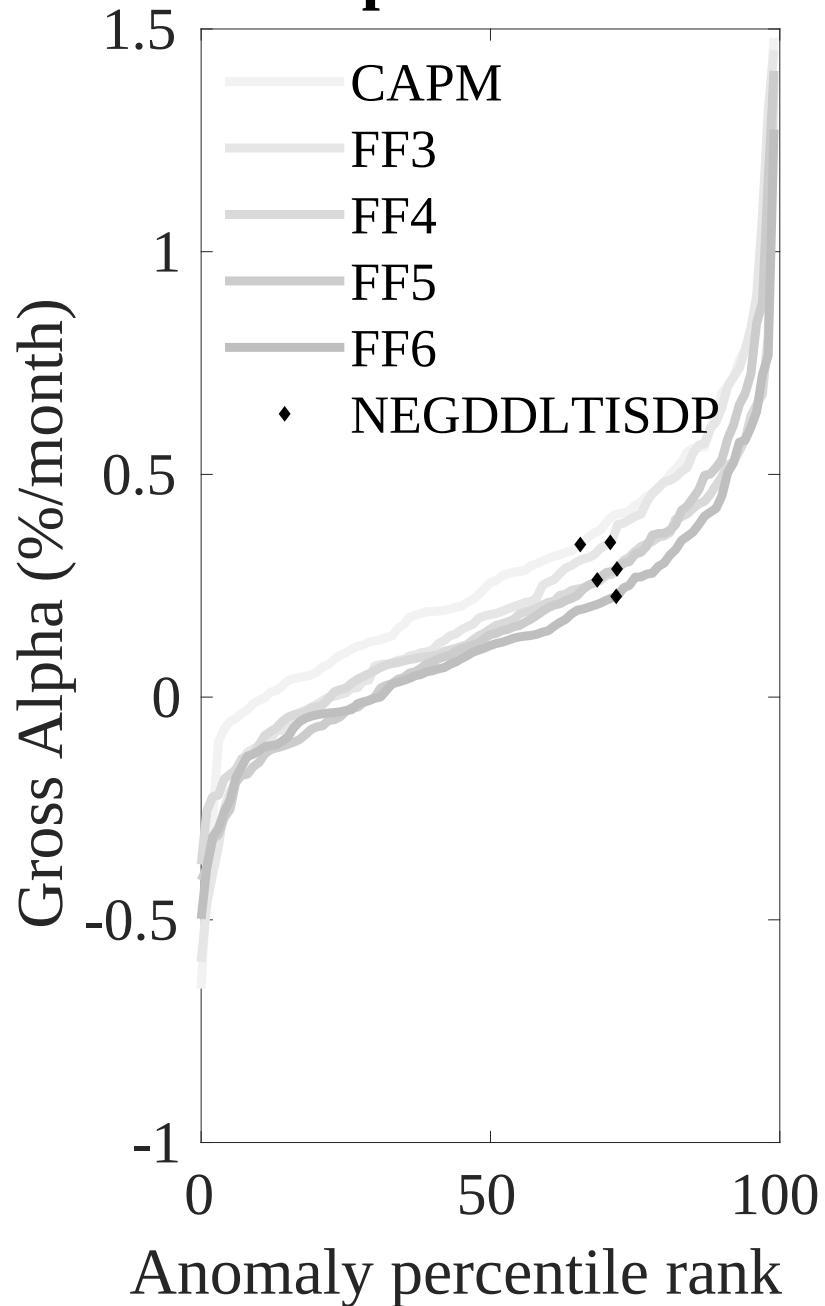


Figure 3: Dollar invested.

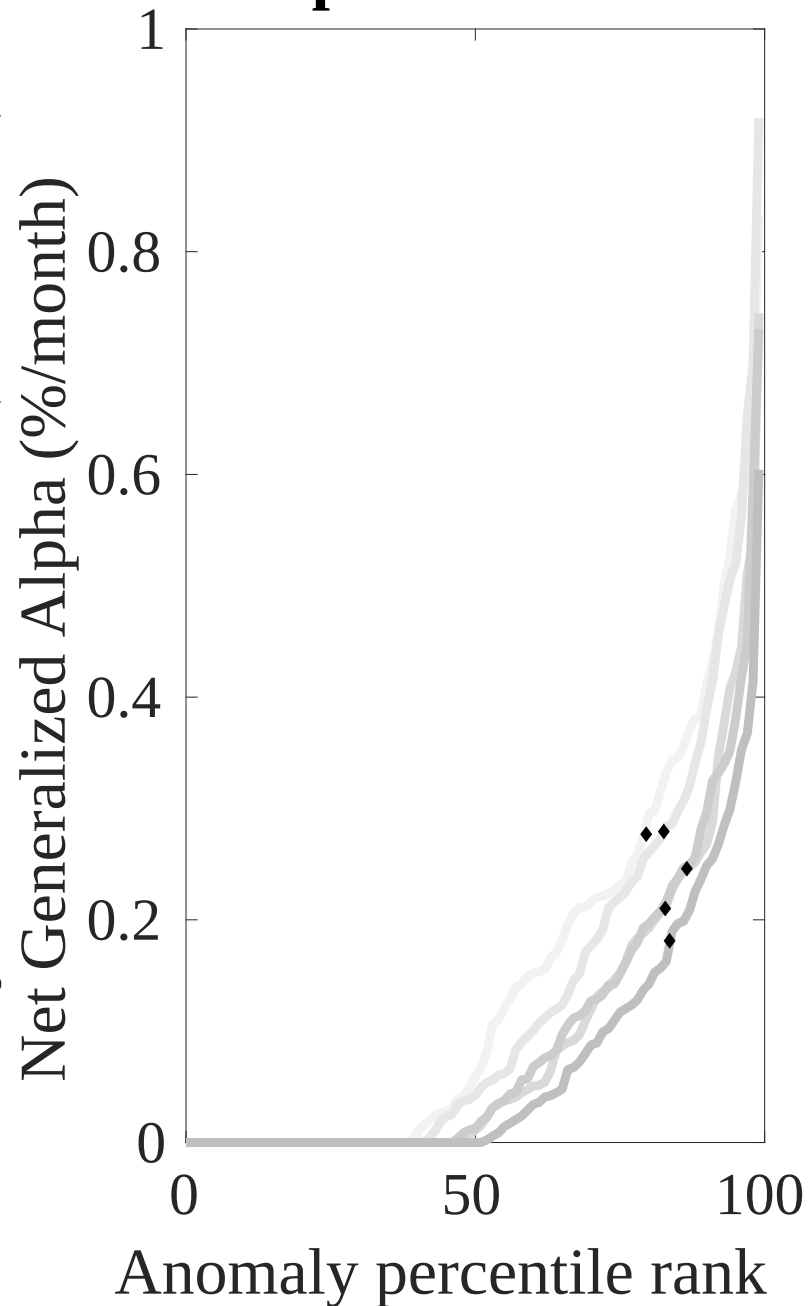
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



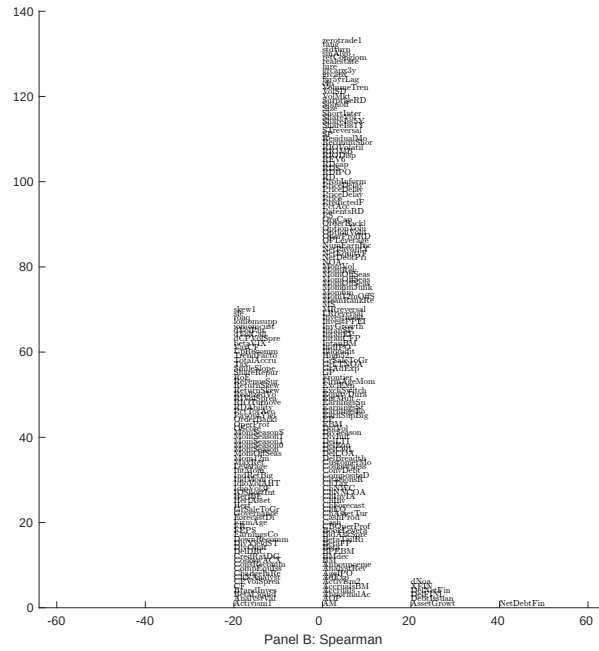
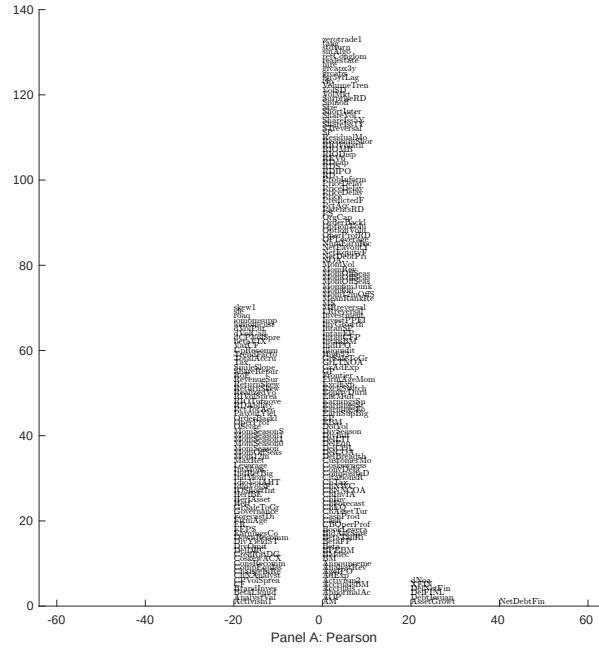


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

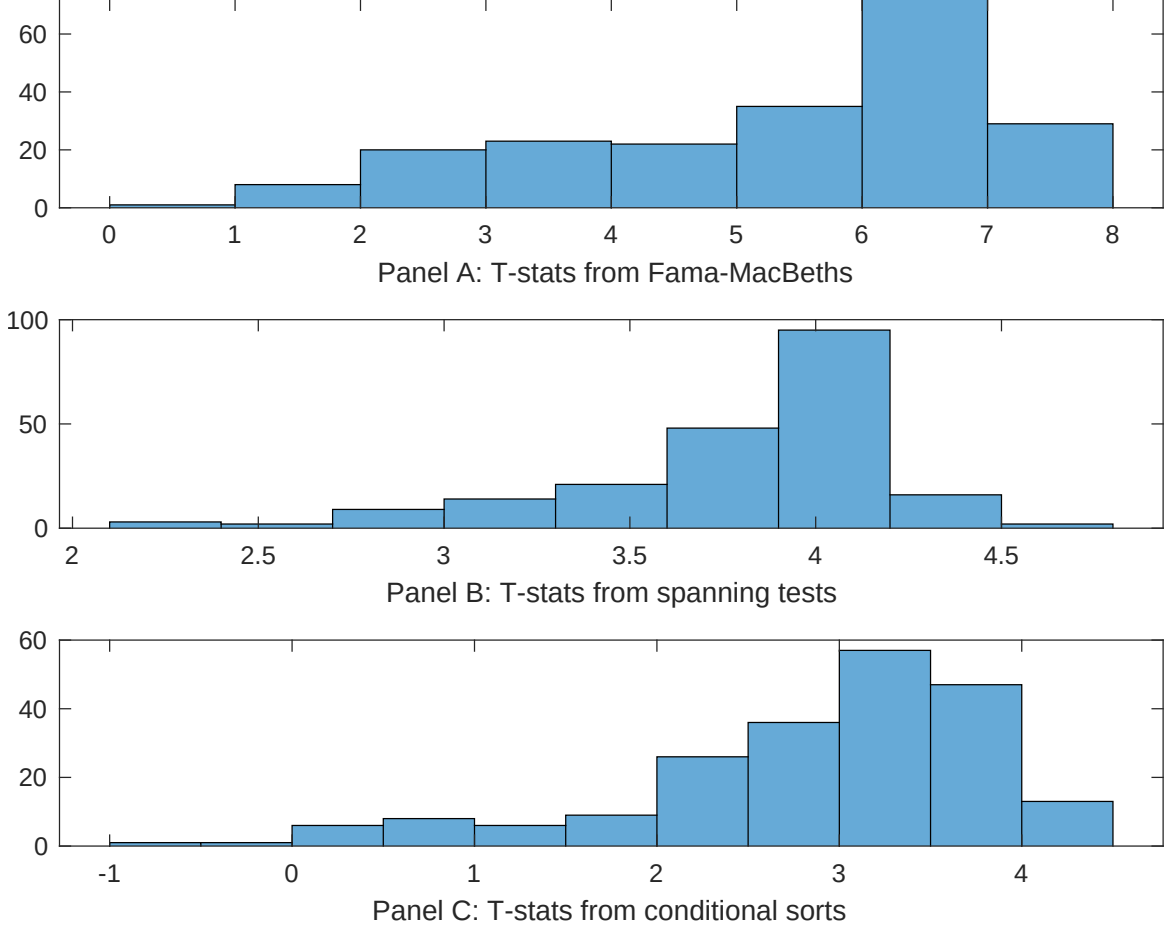


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DFP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.39]	0.13 [5.36]	0.14 [5.72]	0.14 [5.82]	0.14 [5.31]	0.13 [5.37]	0.15 [5.65]
DFP	0.46 [2.32]	0.58 [2.70]	0.76 [3.65]	0.38 [1.79]	0.12 [5.37]	0.11 [5.59]	0.16 [0.62]
Anomaly 1	0.17 [9.28]						-0.58 [-1.26]
Anomaly 2		0.20 [8.70]					0.93 [1.38]
Anomaly 3			0.19 [6.23]				0.10 [1.74]
Anomaly 4				0.11 [8.47]			0.73 [3.52]
Anomaly 5					0.40 [6.88]		0.93 [1.68]
Anomaly 6						0.92 [6.05]	0.92 [0.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DFP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.88]	0.20 [2.95]	0.20 [2.93]	0.21 [3.12]	0.22 [3.14]	0.22 [3.10]	0.18 [2.66]
Anomaly 1	18.20 [4.74]						9.43 [1.85]
Anomaly 2		20.55 [5.58]					12.21 [2.40]
Anomaly 3			11.88 [3.47]				8.87 [2.42]
Anomaly 4				0.78 [0.19]			-3.74 [-0.88]
Anomaly 5					1.83 [0.68]		1.44 [0.52]
Anomaly 6						0.28 [0.08]	-1.51 [-0.39]
mkt	-5.11 [-3.27]	-5.49 [-3.55]	-3.91 [-2.38]	-5.55 [-3.49]	-5.58 [-3.51]	-5.54 [-3.48]	-4.14 [-2.52]
smb	3.69 [1.54]	3.74 [1.58]	9.37 [3.58]	5.51 [2.28]	5.76 [2.38]	5.58 [2.31]	6.68 [2.41]
hml	-13.28 [-4.40]	-13.65 [-4.57]	-13.16 [-4.31]	-14.49 [-4.72]	-14.52 [-4.74]	-14.51 [-4.61]	-11.97 [-3.85]
rmw	-0.86 [-0.28]	-1.48 [-0.48]	-6.32 [-1.71]	0.78 [0.25]	1.05 [0.33]	0.79 [0.25]	-6.59 [-1.77]
cma	26.40 [5.57]	27.17 [5.86]	24.74 [4.84]	31.52 [4.70]	30.84 [5.91]	32.21 [5.65]	25.00 [3.56]
umd	4.42 [2.79]	4.67 [3.00]	5.91 [3.78]	6.02 [3.81]	5.81 [3.63]	5.98 [3.75]	4.20 [2.61]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	19	20	17	16	16	16	20

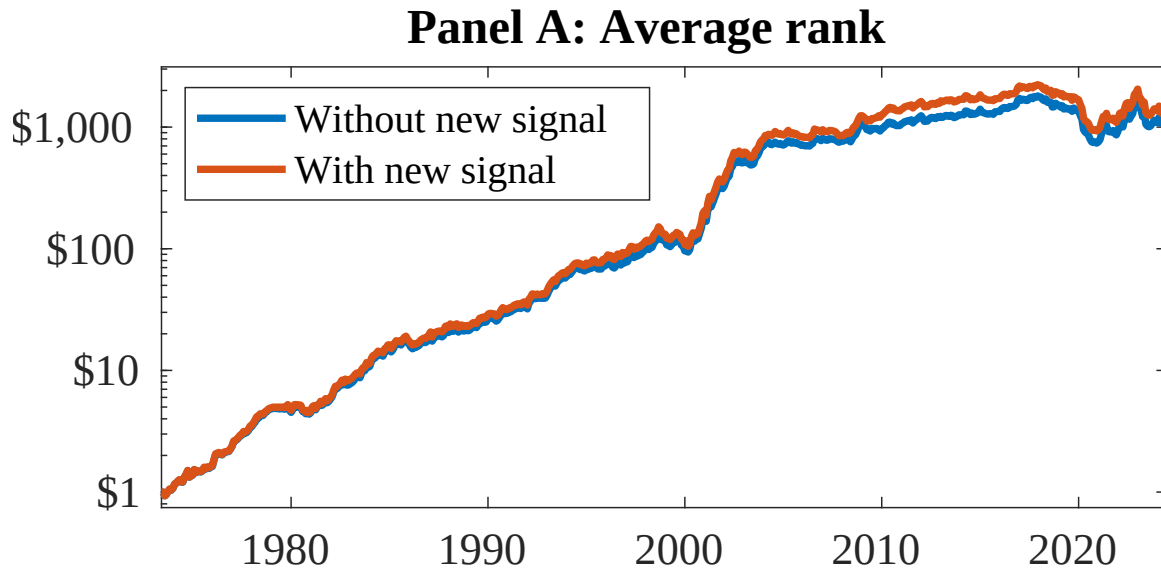


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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